Research Question & Motivation

Question

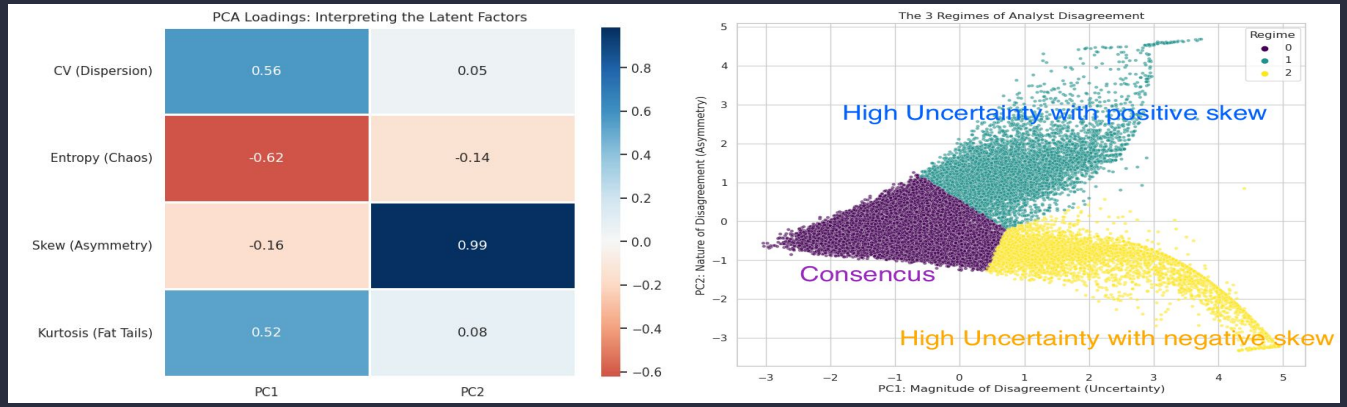
Does greater disagreement in the forecast lead to larger shifts in the price?

- Explores the relationship between the level of uncertainty among analysts and the market volatility after an earnings announcement

Motivation

- Analyze if there is any relationship between pre-earning uncertainty and post-earning price volatility
- Test whether analyst disagreement can be used as a profit-taking signal before earnings

Approach & Key Insight



Data Cleaning & Pre-processing

- Drop unreasonable rows (negative revenue, 0 trading volume, no stock price available)

- Define response variable = abs log return of stock before vs. after

- Compute coefficient of variation (s.d. / mean), apply Yeo-Johnson transformation and standard scaling (prepare for PCA)

Data Visualization

- Distribution check for features after transformation (address non-normality)

- Sector Breakdown: whether disagreement varies across industries

PCA

- Dimension reduction, select features for clustering

- Apply PCA on transformed shape variables, start with four components

-Variance explained: PC1 (47%) & PC2(24%)

- PC1: dispersion-related (CV, kurtosis, entropy) vs. PC2: asymmetry-related (skewness)

K-means Clustering

- Elbow method to determine the optimal number of clusters: 3 clusters

- Apply on PCA selected features: three regimes

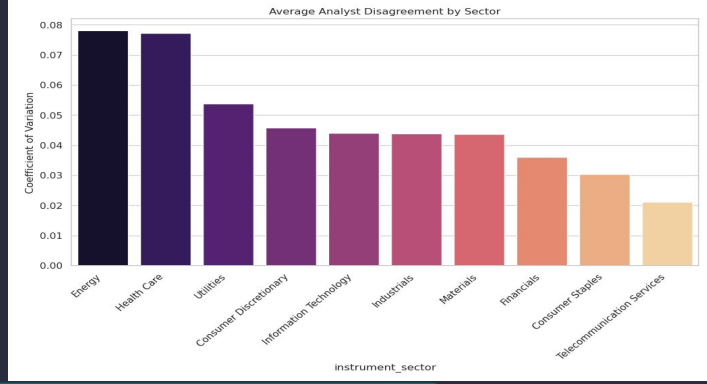
Linear Regression and Random Forest

- Evaluate how well forecast shape matrics explain the magnitude of price movement

- Features: transformed CV, skewness, kurtosis and entropy

- 80/20 split and evaluate out-of-sample R^2

Results, Interpretation & Limitations



Results

Analyst disagreement varies across industries

- Health Care and Energy exhibit higher disagreement
- Consumer Staples and Telecommunication have more clustered estimates

Not all disagreement causes large price move

- Cluster 2 vs. Cluster 3: both high uncertainty but the return of cluster 2 are similar to Consensus group

Low predictive power of linear regression and random forests

- Linear regression R^2 = 0.0521
- Random forests R^2 = 0.0734

Limitations

Low Predictive & Explanatory Power

Other drivers of price movement not modeled

Rely on transformed data (Yeo-Johnson) - may obscure the direct magnitude of price movement in real-world